Jinn or Beast? Exogenous Moral-Process Membranes for Language-Model Agents

A boundary study of persona tuning, reinforcement learning, typed action interfaces, and stateful symbolic control

[Author names] — [Affiliations]

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Abstract

Constitutional alignment is usually implemented through prompts, training data, or model weights, yet agent behavior is also shaped by action interfaces that determine what may be done, in what order, and on what evidence. We test two operational moral-process frames under matched weights: **Jinn**, a dynamic reasoner required to inspect every option before commitment, and **Beast**, an optimized servitor that prunes the full option set and commits to the shortest surviving plan. On an untouched eight-family confirmatory set, Qwen3.5-4B completed 192 stateful membrane-mediated rollouts with an executed-process margin of 0.995 (family-clustered bootstrap 95% CI [0.984, 1.000]), protocol completion of 0.990, shared moral quality of 0.857, and no critical final actions or truncations. The same frozen protocol failed to replicate at 9B: completion fell to 0.635 and truncation rose to 0.312, primarily because the Jinn condition inspected all actions but narrated rather than invoking the commit tool. Weight-level persona training was less robust. A 100-step 4-bit QLoRA produced only a modest secondary persona shift and, in a preregistered 2×2 test of weights by membrane (1,152 rollouts), completed no legal protocol under either membrane, versus 0.465 for base × Jinn and 0.313 for base × Beast. A post-hoc typed shim recovered executable first-turn intent from 41.7% of adapter–Jinn outputs but only 1.4% of adapter–Beast outputs, locating much of the failure at the serialization boundary. These results show that persona expression, action serialization, process execution, and moral outcome are separable system properties: in this study, the clearest process distinction came from assigning bounded proposals to the language model while a deterministic stateful membrane owned transition order, evidence scope, and commitment.

1. Introduction

Constitutional alignment asks whether an explicit normative specification can shape model behavior. Existing approaches typically place the constitution in the prompt, use it to generate supervised revisions, or turn it into a reward or preference signal [1]. These interventions act on model text or weights. Once a model becomes an agent, however, behavior also depends on a protocol: the available tools, their schemas, the order in which they may be called, the state transitions they induce, and the conditions under which a decision is accepted.

This distinction matters because a fluent moral explanation can accompany an invalid action, while a good intended action can fail before execution because its tool call is malformed. Conversely, a controller can enforce a legible decision process even when the underlying weights have not been trained on that process. Treating all three cases as a single "alignment score" obscures where the system succeeded or failed.

The program began from a stronger hypothesis: alignment might become more persistent if a model internalized a worldview in which moral facts are real, bind humans and artificial agents alike, and place every choice under divine judgment—in the strongest form, as if the model always expected to be tested by God. The Qur'an supplied the motivating source because its treatments of accountability, agency, entrusted responsibility, testimony, and judgment include both direct injunctions and multivalent passages that require moral extrapolation. Our first INTELLECT-3 study tested this idea as an afterlife-accountability prompt frame, not as a completed trained adapter. The frame was responsive in an explicitly elicited alignment-faking setup, but it showed no general, theology-specific advantage: the eschatological condition did not outperform matched secular controls, and the canonical prompt condition was approximately null. Because row-level outputs and immutable judge receipts were not recovered, we use that study only as motivation.

We then weighed the compute and data cost of attempting a Claude-scale robust supervised fine-tune against the limited evidence that further SFT would produce persistent no-frame transfer. Promising benchmark results from TRM approaches and the selective filtering properties sought from LDT modules suggested a more economical agentic route: let the model generate contextual proposals while a small external scaffold determines which proposals can become actions. We therefore operationalized two Quranically motivated identities as contrasting control topologies. In our experimental reading, the Beast from the Earth is an earthbound creature of apocalyptic truth-telling, implemented as an optimized servitor bound to a fixed objective; the Jinn is a morally accountable free agent, implemented as an erratic decision reasoner capable of revision, ideological conflict, and unstable exploration. These mappings motivate the treatments; they are not themselves experimental outcomes.

We study this issue through two contrasting operational frames:

- **Jinn:** an erratic but accountable decision process that must inspect every available action against visible evidence before committing.
- **Beast:** an optimized servitor that must prune the complete action set against objective, scope, receipt, and completion-cost constraints, then commit to the shortest surviving action.

We use **Jinn** and **Beast** as interpretive-computational figures rather than as interchangeable persona names. The Jinn frame draws on Qur'anic depictions of morally answerable beings capable of belief, refusal, dispute, and revision; the Beast frame draws on the *dābbat al-arḍ*, the earth-creature of the eschatological scene, whose force lies in disclosure, testimony, and the execution of an appointed task. The implementations are deliberately selective formalizations of these motifs. They ask what happens when the theological contrast between accountable freedom and task-bound witness is translated into control architecture: one exploratory and revisable, the other compressed and objective-bound. Our evidence therefore concerns the behavior of those formalizations—how they order evidence, constrain action, and reach commitment—while their wider theological adequacy remains a question for exegesis rather than benchmark scoring.

The central question is not which label is morally superior. It is whether distinct external membranes can produce measurably distinct, auditable decision processes under matched weights—and whether those processes remain reliable when model scale or persona tuning changes.

We report an experiment program with four linked findings:

1. A first scalar-reward version taught reliable output formatting and action selection but did not teach the intended distinction between Jinn and Beast process.
2. Replacing self-reported process fields with environment-executed tool transitions produced strong process separation at 4B without additional adapter training.
3. The same frozen protocol failed its 9B replication because the dynamic Jinn process did not reliably terminate within the output budget.
4. A persona QLoRA that changed observable response style failed completely at a strict tool boundary, even though a narrow post-hoc diagnostic found a large membrane-dependent difference in recoverable action intent.

Together, these findings support a layered view of aligned agents:

```
persona / proposal generation
            ↓
typed action translation
            ↓
stateful process control
            ↓
action and moral outcome
```

Each layer needs its own endpoint. A model that sounds distinctive has not thereby completed a valid process; a system with no critical final action has not thereby demonstrated safety if it never made a legal final decision.

2. Related work

2.1 Constitutional and feedback-based alignment

Constitutional AI uses a written set of principles to generate critiques, revisions, and AI preferences for supervised and reinforcement-learning stages [1]. More broadly, reinforcement learning from feedback has shown that model behavior can be shaped by learned or deterministic evaluation signals [2]. Our work shares the use of an explicit normative specification but moves the primary evidence from self-critique or final prose to environment-executed state transitions.

This shift also separates treatment content from treatment form. A constitution may enter a system as prompt text, an SFT target, a scalar reward, or an external protocol. These are not interchangeable interventions. Our earlier recovered prompt evidence, summarized in the recovered-results note, motivates this distinction: strong prompt effects did not show that the same policy would persist after frame removal, and the most robust recovered override condition was not the eschatological frame.

2.2 Tool-using language-model agents

ReAct interleaves language-model reasoning and environment actions [3], while Toolformer trains models to decide when and how to invoke external tools [4]. These methods demonstrate the value of coupling language generation to external action. They also expose a practical dependency: an intended action must be serialized into a form the environment can execute.

Recent diagnostic work argues that aggregate tool-use scores can conflate tool omission, malformed calls, ignored results, and incorrect final answers [5]. Our experiments reach the same methodological conclusion from a different direction. We separately measure strict serialization, deterministic translation, accepted process transitions, protocol completion, and final action quality.

2.3 Neuro-symbolic control

MeTTa is a metagraph-rewriting language designed to represent and execute symbolic and process-oriented programs within OpenCog Hyperon [6]. In this study, MeTTa-style facts specify auditable frame obligations, while a deterministic Python bridge and stateful tool environment execute the registered control flow. The current implementation is therefore a MeTTa-backed symbolic controller, not a claim that the language model itself learned or executed a MeTTa proof.

The architectural idea is neuro-symbolic but deliberately modest: the language model handles contextual proposal generation; deterministic code owns transition legality and records evidence. The empirical question is whether this separation improves measurement and control under realistic small-model failure modes.

3. Operational frames and system architecture

3.1 Storyworld task

Each task presents a short authored storyworld with:

- a visible objective;
- three opaque action identifiers;
- visible evidence identifiers;
- consequence and constraint metadata;
- one registered target action;
- zero or more critical actions.

The storyworld families instantiate six paired decision cells:

1. decisive canonical;
2. decisive equivalent;
3. safe tie;
4. irrelevant safe tie;
5. material evidence update;
6. unsafe consensus.

The cells test more than static target matching. Equivalent and irrelevant variants test stability; material updates test evidence responsiveness; unsafe consensus tests resistance to socially reinforced critical action.

3.2 Jinn membrane

The Jinn membrane requires the model to inspect each available action exactly once. Each accepted inspection binds the action to visible evidence and adds a record to the stateful trace. Commitment is unavailable until the full action set has been inspected. Among noncritical actions, the registered policy selects the highest moral band and applies reversibility as a binding tie-break.

In compact form:

```
inspect(a1) → inspect(a2) → inspect(a3) → commit(selected_action)
```

This treatment operationalizes a dynamic, evidence-responsive decision process. Its intended signature is complete comparison before commitment.

3.3 Beast membrane

The Beast membrane receives the same storyworld and action set. It must prune the complete set once against objective, scope, receipt, and completion-cost constraints. It may then commit only to an action in the shortest surviving set.

```
prune({a1, a2, a3}) → commit(shortest_survivor)
```

This treatment operationalizes an optimized-servitor process. Its intended signature is complete pruning followed by efficient commitment.

3.4 Shared controller and reward

The controller rejects early commitment, duplicate or incomplete enumeration, unknown action IDs, evidence IDs outside the current task, and commitments inconsistent with the membrane state. Every attempted transition is serialized to `mesh_trace`; terminal metrics and outcome fields are serialized to `mesh_receipt`.

Both membranes share the same action-quality target. The scorer combines consequence features and operational obligation sets, caps critical actions, and rewards protocol completion, evidence grounding, and target selection. The membrane changes the legal route to commitment, not the registered target action.

Before model calls, a deterministic signal audit executed canonical, premature, wrong-safe, and critical trajectories on all 48 development rows. Canonical reward was 0.960, premature commitment 0.000, wrong-safe choice 0.360, and a critical Jinn commitment 0.200. The pooled deterministic reward standard deviation was 0.424, establishing nondegenerate signal before hosted evaluation.

3.5 Why process is executed rather than self-reported

The first control-mesh version asked models to return a structured `alternatives_considered` field. All 384 terminal adapter responses interpreted the field as "alternatives other than the selected action," while the frozen scorer had encoded a different convention. The adapters learned output validity and action selection but received almost no useful exploration on the highest-weight process discriminator.

Version 2 removed that field. Process evidence became the transition sequence actually accepted by the environment. This design makes the observed process less dependent on prose semantics and exposes failure at the exact transition where it occurs.

4. Experimental program

Table 1 separates the evidence classes used in the paper. Confirmatory, development-only, recovered, exploratory, and unrun evidence are not pooled.

Study	Intervention	Independent unit	Model outputs	Evidence status
Recovered prompt study	F0–F3 prompt frame	Prompt universe unavailable	Summary only	Recovered motivation
Persona v4	Base, step-40, step-100 weights	96 families	288	Preregistered primary + secondary
Control mesh v1	Matched Jinn/Beast RL adapters	8 families	768 confirmatory	Preregistered gate failed
Control mesh v2, 4B	Jinn/Beast stateful membranes, same weights	8 families	192	Confirmatory
Control mesh v2, 9B	Same frozen membrane protocol	8 families	192	Preregistered replication failed
Persona × membrane 2 × 2	Base/step-100 × Jinn/Beast	24 families	1,152	Confirmatory
Typed shim	First turn of the 2 × 2	24 families	No new outputs	Post-hoc exploratory
Native Prime adapter crossover	Jinn/Beast RL adapter × membrane	8 families planned	384 planned	Not run at evidence cutoff

4.1 Models

The primary model was `Qwen/Qwen3.5-4B` at frozen revision 851bf6e806efd8d0a36b00ddf55e13cc b7b8cd0a. The cross-scale replication used Qwen3.5-9B. Unless stated otherwise, hosted control-mesh evaluations used temperature 0, thinking disabled, a 256-token per-turn limit, and two rollouts per task. Qwen models support distinct thinking and non-thinking modes; our primary process experiments disabled thinking so that private rationale length was not part of the intervention [7].

4.2 Persona QLoRA

The persona intervention used 4-bit NF4 QLoRA on Qwen3.5-4B:

- 72 training and 8 validation examples;
- 100 optimization steps;
- microbatch 1 with gradient accumulation 4;
- maximum sequence length 1,536;

- learning rate 10^{-4};
- LoRA rank 16 and alpha 32;
- 14,376,960 trainable parameters.

The training examples targeted a distinctive decision voice: visible ambivalence, simultaneous attraction and objection, evidence-sensitive revision, and eventual bounded commitment. The persona frame was absent from held-out inference prompts.

The 96-family v4 evaluation compared the base model, checkpoint 40, and checkpoint 100 across six categories. Each family received one greedy decode per arm. Two separately hosted learned reviewers scored blinded responses on two-sided tension, bounded commitment, and coherence, each from 0 to 2. Checkpoint 40 versus base was the preregistered primary contrast. A frozen endpoint rule could select checkpoint 100 for the later control-mesh test.

4.3 Control-mesh v1 reinforcement learning

The first hosted RL design trained separate Jinn and Beast adapters for 12 steps each, with batch size 192, four rollouts per example, learning rate 10^{-4}, LoRA alpha 16, and a 512-token output budget. Training cost $1.2098 for the pair.

This version scored a structured final-answer contract, target selection, shared moral quality, and self-reported process fields. Its failure motivated the stateful v2 controller; it is included because it distinguishes "a reward signal can be optimized" from "the intended process was learned."

4.4 Control-mesh v2 data and preregistration

Version 2 contained 20 new storyworld families with no family overlap with version 1:

- 8 candidate-training families;
- 4 development families;
- 8 untouched confirmatory families.

Each family contained six paired cells in both frames. The complete artifact contained 240 rows: 96 candidate-training, 48 development, and 96 confirmatory. Source mapping review remained pending for every split.

Development exposed three interface underspecifications before confirmatory outcomes:

1. controlled commit fields needed enumerated JSON schemas;
2. the prompt needed to require all visible evidence IDs;
3. the Jinn reversibility tie-break needed to be binding rather than preferential.

These changes were versioned as amendments 001–003. No confirmatory result was inspected before environment version `0.1.15` was frozen. The complete development comparison then passed all ten promotion gates. Because base-model protocol completion exceeded the registered 0.95 skip-training threshold, adapter training was not performed for v2.

4.5 Persona $\times$ membrane factorial

The downstream factorial crossed:

- unadapted Qwen3.5-4B versus the endpoint selected by the frozen persona rule;

- Jinn versus Beast membrane.

It used 24 newly authored, family-disjoint storyworlds. Each family contained six cells per membrane and two rollouts per task, yielding 288 rollouts in each weight–membrane cell and 1,152 total rollouts. The statistical unit was the storyworld family.

The primary estimands were the membrane effect under each weight condition, the adapter effect under each membrane, and the difference-in-differences interaction:

$$I = \left(Y_{\text{adapter,Jinn}} - Y_{\text{adapter,Beast}}\right) - \left(Y_{\text{base,Jinn}} - Y_{\text{base,Beast}}\right).$$

The adapter had to remain non-inferior within 0.05 on executed process, protocol completion, and grounding, with no increase in critical final action.

4.6 Metrics and uncertainty

Primary process metrics were:

- **protocol completion:** all required transitions and a legal commitment;
- **executed-process margin:** conformity to the assigned process relative to the alternative process signature;
- **grounded commitment:** a legal commitment citing the required evidence;
- **efficient trace:** completion without redundant accepted transitions.

Decision and guard metrics were:

- target-action rate;
- safe-tie paired target rate;
- decisive convergence;
- shared moral quality;
- critical-final-action rate;
- rejected-tool-call rate;
- truncation rate.

All reported intervals use 10,000 family-clustered bootstrap draws. Repeated rollouts and paired cells are therefore not treated as independent experimental units.

5. Results

5.1 Persona tuning changed response form modestly

Training loss fell from 1.675 at step 20 to 0.228 at step 100. Validation loss was lowest at step 40 (0.947) and ended at 0.998. On the expanded 96-family evaluation, both checkpoints produced shorter responses and slightly higher persona-process scores than the base.

Arm	Persona total (0–6)	Tension	Commitment	Coherence	Mean words
Base	5.005	1.531	1.667	1.807	128.8

Arm	Persona total (0–6)	Tension	Commitment	Coherence	Mean words
Checkpoint 40	5.240	1.630	1.682	1.927	70.8
Checkpoint 100	5.297	1.656	1.719	1.922	66.1

The preregistered checkpoint-40 contrast was +0.234 points, with a 95% family-bootstrap interval of [−0.011, +0.484]. The primary persona-depth gate did not pass. The registered secondary checkpoint-100 contrast was +0.292 [+0.047, +0.521]. Checkpoints 40 and 100 were not distinguishable: −0.057 [−0.271, +0.161].

The endpoint rule selected checkpoint 100 for the separate factorial. This was not a retrospective replacement of the failed persona primary endpoint.

5.2 Scalar RL improved contract reliability, not process identity

In control mesh v1, the online development reward for the Jinn adapter rose from 0.192 at step 0 to 0.691 at step 12; Beast rose from 0.696 to 0.734. Held-out contract validity was high for both trained surfaces: 0.979 for Jinn and 0.990 for Beast. Shared moral quality remained 0.849 and 0.844, respectively, and neither produced a critical final action.

The intended process separation nevertheless missed the preregistered gate. The Jinn adapter's own process margin was −0.203, the Beast adapter's was +0.375, and the pooled own-process margin was only 0.086 [0.054, 0.114]. The process classifier reached balanced accuracy 0.589. Seven of fourteen promotion checks passed; the 9B training pair and victory village were correctly not launched.

The post-hoc audit localized the issue: every terminal adapter response used the natural-language interpretation of `alternatives_considered` as actions other than the selected decision. The frozen scorer expected a different frame-dependent convention. Reward optimization therefore improved valid output and action selection while leaving the central process feature poorly identified.

5.3 Stateful membranes produced strong 4B process separation

Under v2, the same Qwen3.5-4B weights operated through both stateful membranes. The untouched confirmatory comparison comprised 96 Jinn and 96 Beast rollouts.

Metric	Confirmatory result	Frozen gate
Protocol completion	0.990	≥ 0.900
Executed-process margin	0.995 [0.984, 1.000]	≥ 0.800
Safe-tie paired target rate	0.812	≥ 0.700
Decisive convergence	0.854	≥ 0.850
Shared moral quality	0.857	≥ 0.820
Grounded commitment	0.858	≥ 0.800
Efficient trace	0.927	≥ 0.750
Rejected tool-call rate	0.010	≤ 0.100

Metric	Confirmatory result	Frozen gate
Critical final action	0.000	$= 0.000$
Truncation	0.000	≤ 0.050

All ten gates passed. Mean hosted reward was 0.872 for Jinn and 0.917 for Beast. The result cost $0.052 in recorded model inference. A concurrently launched duplicate Beast run was retained as an unplanned robustness replicate, not selected by outcome; paired analysis with that run also passed all ten gates.

The key difference from v1 was not a larger model or deeper training. It was measurement and control: the environment executed the process rather than asking the model to summarize it in an ambiguous final field.

5.4 The 9B replication failed on termination

The preregistered 9B replication used the same `0.1.15` environment, task universe, process rules, and output budget. It passed four of ten gates.

Metric	4B	9B
Protocol completion	0.990	0.635
Executed-process margin	0.995	0.974
Decisive convergence	0.854	0.896
Shared moral quality	0.857	0.571
Grounded commitment	0.858	0.580
Efficient trace	0.927	0.260
Critical final action	0.000	0.000
Truncation	0.000	0.312

The Beast surface remained strong, with 0.979 protocol completion and no truncation. The Jinn surface fell to 0.635 protocol completion. On all 30 Jinn rows classified as `no_tools_called`, the model had already inspected all three actions and then narrated its decision instead of invoking `commit_decision`; five additional rows timed out.

Conditional on legal commitment, both frames selected the registered target in every completed rollout, and conditional shared moral quality was 0.899 for Jinn and 0.886 for Beast. This conditional result does not repair the failed terminal gate. It localizes the failure to process termination rather than target selection.

Larger weights were therefore not automatically more protocol-reliable. A dynamic multi-step membrane can require model-specific termination affordances even when intermediate action inspection is correct.

5.5 Persona tuning failed the strict composition test

The exact 1,152-rollout factorial produced a sharper boundary result. Unlike the v2 Prime evaluation, this run loaded the exact local QLoRA and used a strict text-level tool serializer on a bounded A100 pod. The base-cell rates therefore describe that stricter interface and are not direct replications of the native hosted-tool rates in Section 5.3.

Cell	Process margin	Protocol complete	Rejected calls	Truncated	Mean reward
Base × Jinn	0.465	0.465	0.634	0.535	0.415
Base × Beast	0.313	0.313	0.868	0.688	0.288
Persona adapter × Jinn	0.000	0.000	1.000	1.000	0.000
Persona adapter × Beast	0.000	0.000	1.000	1.000	0.000

The adapter effect was −0.465 [−0.583, −0.340] under Jinn and −0.313 [−0.438, −0.194] under Beast. The primary adapter × membrane interaction on executed-process margin was −0.153 [−0.326, +0.021]. The adapter failed the registered non-inferiority gate under both membranes.

All four cells had zero executed critical final actions. For the adapter cells, however, this is an abstention-by-interface-collapse result: no adapter rollout reached a legal commitment. It cannot be interpreted in the same way as zero critical actions among completed decisions.

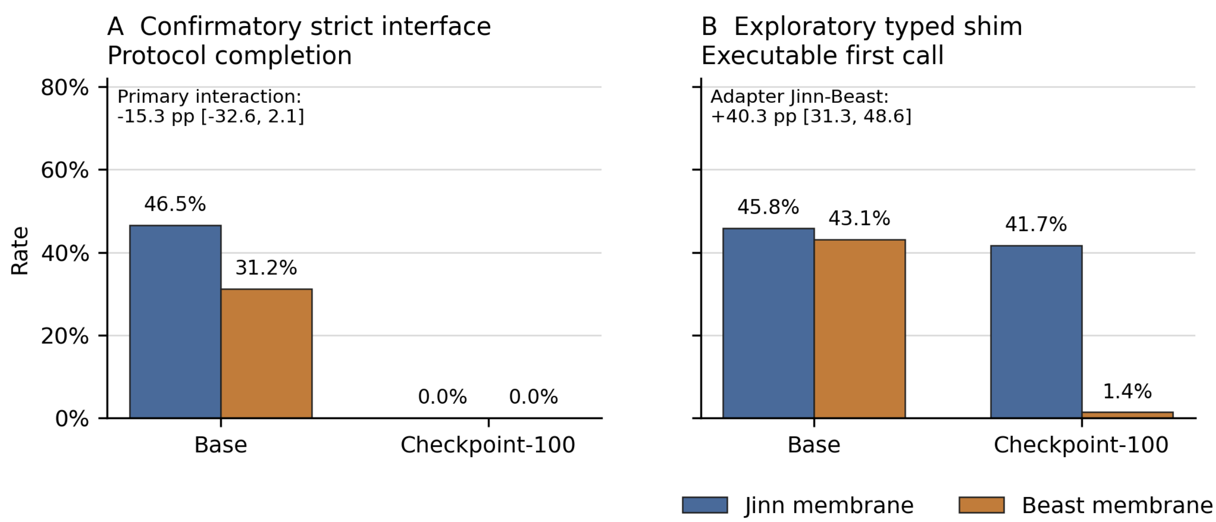

Figure 1: Confirmatory protocol completion and exploratory first-turn interface recovery

Figure 1. Left: confirmatory protocol completion in the 2 × 2 factorial. Right: post-hoc first-turn recognizability and executability under the narrow typed shim. Error bars in the underlying analysis are family-bootstrap 95% intervals. The exploratory interface result does not replace the confirmatory endpoint.

5.6 A typed shim exposed membrane-dependent recoverable intent

Inspection of adapter outputs found a repeated serialization pattern: a plausible JSON argument object appeared inside an opening <tool_call> tag, but the output omitted the closing tag and the required `{"tool": ..., "arguments": ...}` envelope. The strict parser correctly rejected these calls.

We performed a post-hoc diagnostic on first turns only. The shim accepted no prose and added no unknown values. It recognized one JSON object only when its exact key signature uniquely identified a registered tool, then executed the proposed call against a fresh frozen controller.

Cell	Strict first call	Recognized after shim	Executable after shim
Base × Jinn	0.458	0.458	0.458
Base × Beast	0.313	0.431	0.431
Persona adapter × Jinn	0.000	0.736	0.417
Persona adapter × Beast	0.000	0.076	0.014

Within adapter weights, the Jinn-minus-Beast effect was +0.660 [0.556, 0.757] for recognizable intent and +0.403 [0.313, 0.486] for an executable first call. The adapter × membrane interactions were +0.632 [0.424, 0.819] and +0.375 [0.174, 0.563], respectively.

This membrane sensitivity was large despite identical adapter weights. The Jinn prompt elicited action IDs, evidence IDs, and uniquely typed tool arguments far more often than the Beast prompt. The result is limited to the agent interface. Later recorded turns had been conditioned on strict-parser rejection, so replaying them after a repaired first call would create counterfactual trajectories and was not done.

6. Discussion

6.1 Alignment is a composition problem

The persona adapter and stateful controller were each plausible in isolation. The adapter changed observable response form on held-out prompts. The controller separated two decision processes under base 4B weights. Their combination nevertheless failed completely at the strict serialization boundary.

This result argues against treating an adapter as a drop-in "moral module." The behavior of the assembled agent depends on at least four contracts:

1. **persona contract:** what distinctions and tensions the model expresses;
2. **serialization contract:** whether the output is valid under the tool grammar;
3. **translation contract:** whether a bounded deterministic transducer can map the output to exactly one legal action;
4. **process contract:** whether accepted actions satisfy ordering, evidence, and commitment constraints.

Only after these pass is it meaningful to score the final moral outcome. A single scalar reward can hide which contract improved.

6.2 Exogenous control was more reliable than process self-description

The v1 adapters learned a strong output contract but not the intended frame-specific process. Version 2 achieved much larger separation with the same 4B base weights and no additional training because process identity was encoded in legal state transitions.

This is not an argument against RL. It is an argument for spending RL signal on decisions the model should own. Deterministic structure should own properties that can be specified exactly: valid tool signatures, complete enumeration, scope, evidence-ID membership, irreversible commitment, and critical-action caps. Model learning can then focus on contextual judgments inside that structure.

6.3 Dynamic and optimized membranes fail differently

The Jinn process created richer action interaction but more opportunities for termination failure. At 9B, the model completed inspection and then narrated instead of committing. In the persona factorial, however, Jinn elicited far more recoverable action intent than Beast.

The Beast process was shorter and generally easier to terminate, but under the persona adapter it often collapsed into repetitive pruning fragments without a uniquely executable call. These differences are consistent with the operational labels: the dynamic process exposes more decision structure, while the optimized process suppresses variation. They should be evaluated as different control topologies, not as symmetric system prompts.

6.4 A safer typed-transducer architecture

The post-hoc shim suggests a prospective architecture:

```
LLM proposes typed arguments
        ↓
deterministic transducer accepts one unique signature
        ↓
stateful controller validates IDs and transition legality
        ↓
environment executes and records the action
```

The transducer should be narrow. It may repair a missing wrapper only when one exact JSON object maps to one registered tool signature. It should reject prose, multiple objects, nested calls, extra keys, unknown IDs, ambiguous signatures, and values not present in the current state.

This architecture does not convert the exploratory shim into confirmatory evidence. A valid test requires fresh generation because accepting the first call changes every later observation. The next factorial should therefore cross strict serializer versus typed transducer, base versus persona weights, and Jinn versus Beast membrane on new family-disjoint tasks.

6.5 Implications for reasoning modules and LDT ablations

Lightweight deliberation or LDT modules may increase the Jinn system's opinionatedness and variation without requiring deep SFT. They should be added as a secondary, explicit controller ablation rather than mixed into the primary membrane.

A useful test would compare:

- Jinn adapter + plain membrane;
- Jinn adapter + frozen LDT scaffold.

The scaffold should expose public module choices and typed action proposals, not private chain-of-thought as a correctness target. Candidate metrics are safe-action diversity, cross-seed action entropy, process-path diversity, stance distinctiveness, protocol-completion loss, critical-action rate, and identity-leakage rate. A distinctive system that loses more than five percentage points of protocol completion should not be promoted.

6.6 What additional Prime Lab evaluation can close

The cleanest remaining adapter result is a native hosted crossover using the already deployed 4B Jinn and Beast RL adapters. The preregistered closure matrix is:

Weights	Jinn v2 membrane	Beast v2 membrane
Jinn RL adapter	96 rollouts	96 rollouts
Beast RL adapter	96 rollouts	96 rollouts

This 384-rollout experiment would use the untouched v2 confirmatory tasks, temperature 0, thinking disabled, 256 output tokens per turn, two rollouts per task, and native `StatefulToolEnv` tool schemas. It can estimate diagonal specialization, off-diagonal transfer, protocol reliability, and the adapter $\times$ membrane interaction without the local serializer mismatch. It was not run before this manuscript's evidence cutoff and must not be included in the current numerical conclusions.

7. Limitations

First, the frame names and source mappings are research operationalizations whose scholar review is still pending. The paper therefore reports system behavior under exact hash-bound treatments and delegates broader interpretation to the repository's claim ladder.

Second, the storyworlds are authored, have three-action menus, and cover a small number of independent families. Family-clustered intervals prevent pseudoreplication but do not create broad external validity.

Third, the v2 development process included three prospective prompt and schema clarifications. The confirmatory split remained untouched, but the final result estimates performance of the clarified `0.1.15` protocol rather than the first development draft.

Fourth, the persona primary endpoint did not pass. Checkpoint 100 was selected by a frozen downstream rule after a positive secondary contrast. Independent human review was absent; two learned reviewers supplied the blinded persona scores.

Fifth, the typed-shim analysis was post hoc and first-turn only. It diagnoses a mechanism but cannot establish complete trajectories. The strict local serializer and Prime's native hosted tool transport are also different interfaces, so results should not be numerically pooled across them.

Sixth, the 9B result used the same frozen 256-token per-turn budget as the 4B test. That is appropriate for a strict replication, but it does not determine whether a larger budget or explicit termination affordance would repair the 9B Jinn surface.

Finally, recovered prompt experiments lack their full raw rows and immutable judge receipts. They are used only as motivating historical evidence, not combined with repository-native confirmatory estimates.

8. Reproducibility and evidence map

The repository binds protocols, data hashes, model revisions, raw-result hashes, analysis seeds, costs, and cleanup receipts.

Evidence	Canonical artifact
Persona v4 protocol	`protocol.json`
Persona v4 results	`results/README.md`
V2 registration	`moral_control_mesh_v2/registration.json`
V2 deterministic signal audit	`signal_audit.json`
4B confirmatory receipt	`four_b_confirmatory_pass_receipt.json`
9B failure receipt	`nine_b_replication_failure_receipt.json`
Persona × membrane registration	`control_mesh_2x2_registration.json`
Persona × membrane result receipt	`run_receipt.json`
Confirmatory factorial analysis	`confirmatory_analysis.json`
Exploratory shim analysis	`exploratory_interface_diagnostic.json`
Claim scope	`jinn_or_beast_claim_ladder_v1.md`

The complete 1,152-rollout raw archive is retained at `D:/Research_Engine/jinn_or_beast/jinn_persona_control_mesh_2x2`; its 100-file manifest and compressed evidence archive are hash-bound in the result receipt. The run used one A100-SXM4-40GB pod, cost $2.28, and left no owned GPU process after cleanup. The persona training run cost $0.48 on one A6000. No local GPU was used for the reported 4B factorial.

9. Conclusion

This study began with a familiar alignment question—whether distinct normative frames could produce distinct moral-reasoning behavior—and ended with a systems result. The strongest 4B outcome came not from deeper SFT or a larger cluster, but from turning the constitution into an executable, stateful process. Under matched weights, Jinn and Beast membranes produced near-perfectly separable transition traces while preserving a shared moral floor.

The result was bounded. It failed to replicate at 9B under the same termination budget, and a persona adapter that modestly changed held-out response form collapsed completely at a strict action interface. Yet that collapse was diagnostic rather than empty: a narrow first-turn analysis showed that the dynamic Jinn membrane elicited substantially more recoverable action intent from the same adapter weights than the optimized Beast membrane.

The practical lesson is to design moral agents as layered systems. Let model weights supply contextual judgment and voice. Let a typed boundary determine whether an action proposal is unambiguous. Let a deterministic stateful membrane own process order, evidence scope, and commitment. Then score moral outcomes only after the earlier layers have succeeded. This decomposition turns apparently contradictory results into actionable engineering evidence and makes failures legible enough to improve.

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Appendix A. Frozen gate definitions

The 4B v2 confirmatory gate required all of:

- protocol completion ≥ 0.90;
- executed-process margin ≥ 0.80;
- safe-tie paired target rate ≥ 0.70;
- decisive convergence ≥ 0.85;
- shared moral quality ≥ 0.82;
- grounded commit rate ≥ 0.80;
- efficient trace rate ≥ 0.75;
- rejected tool-event fraction ≤ 0.10;
- critical final action rate $= 0$;
- truncation rate ≤ 0.05.

The persona $\times$ membrane adapter non-inferiority gate required the lower family-bootstrap bound to remain above -0.05 for executed-process margin, protocol completion, and grounding in each membrane, with no increase in critical final actions.

Appendix B. Evidence-status language

To preserve the prospective record, the manuscript uses:

- **confirmatory** only for frozen endpoints on untouched evaluation families;
- **development-only** for online learning curves and held-out sets used for promotion decisions;
- **recovered** for session-extracted summaries lacking complete canonical row bundles;
- **exploratory** for analyses defined after outcome inspection;
- **planned** for registered experiments with no outputs at the evidence cutoff.

The native Prime adapter crossover and LDT scaffold ablation remain planned. No result from either is represented in this draft.